

(12) **United States Patent**
Rawat et al.

(10) **Patent No.:** **US 12,388,634 B2**
(45) **Date of Patent:** **Aug. 12, 2025**

(54) **CUSTOMER PREMISES EQUIPMENT
ACCESS USING PASSWORD-OF-THE-DAY**

(71) Applicant: **Charter Communications Operating,
LLC**, St. Louis, MO (US)

(72) Inventors: **Deependra Singh Rawat**, Highlands
Ranch, CO (US); **Shlomo Ovadia**,
Denver, CO (US); **Daniel Roy Lynch**,
Englewood, CO (US)

(73) Assignee: **Charter Communications Operating,
LLC**, St. Louis, MO (US)

(*) Notice: Subject to any disclaimer, the term of this
patent is extended or adjusted under 35
U.S.C. 154(b) by 218 days.

(21) Appl. No.: **18/311,697**

(22) Filed: **May 3, 2023**

(65) **Prior Publication Data**

US 2024/0372715 A1 Nov. 7, 2024

(51) **Int. Cl.**
H04L 9/08 (2006.01)

(52) **U.S. Cl.**
CPC **H04L 9/0863** (2013.01); **H04L 2209/56**
(2013.01)

(58) **Field of Classification Search**
CPC **H04L 9/0863**; **H04L 2209/56**
See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

2009/0264137 A1* 10/2009 Soliman H04W 64/00
455/456.1

2014/0281506 A1* 9/2014 Redberg H04L 63/0442
713/159

2018/0337783 A1* 11/2018 Lu H04L 63/083

* cited by examiner

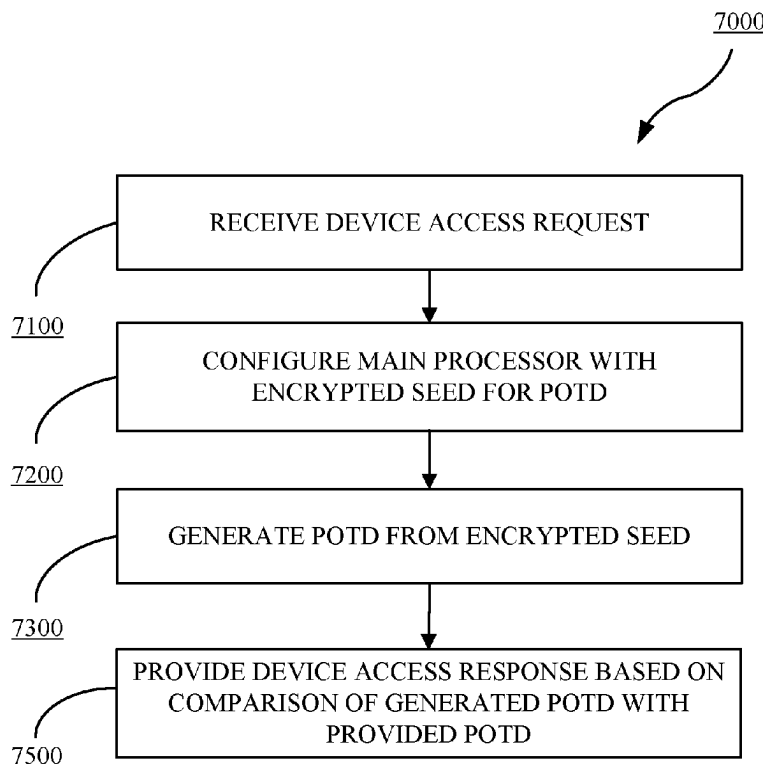
Primary Examiner — Henry Tsang

(74) *Attorney, Agent, or Firm* — Young Basile Hanlon &
MacFarlane, P.C.

(57) **ABSTRACT**

Methods and systems for configuring a main or security processor in a customer premises equipment (CPE) with an encrypted seed for password-of-the-day (PoTD) processing is described. A CPE includes a modem processor, a main processor, and a PoTD component included on the main processor. The main processor receives, via the modem processor from a user device, an access request to the CPE. In response to the access request, the PoTD component accesses an encrypted seed stored on the main processor, generates a PoTD from the encrypted seed, and provides an access response based on comparison of the generated PoTD with a PoTD provided via the user device. A security processor can be used in lieu of the main processor. The PoTD component is then included in the security processor and the encrypted seed is then stored on the security processor.

24 Claims, 10 Drawing Sheets



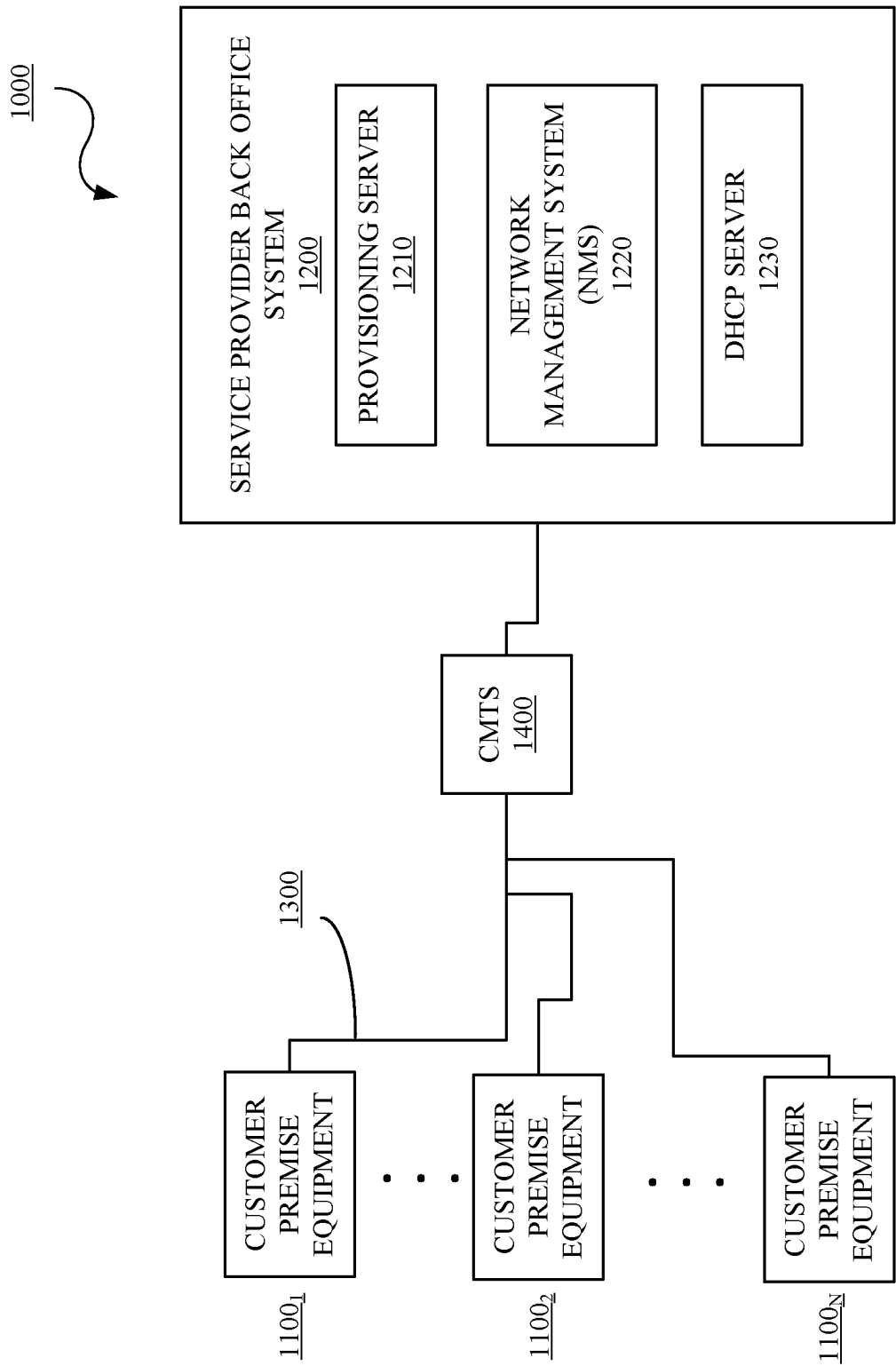


FIG. 1

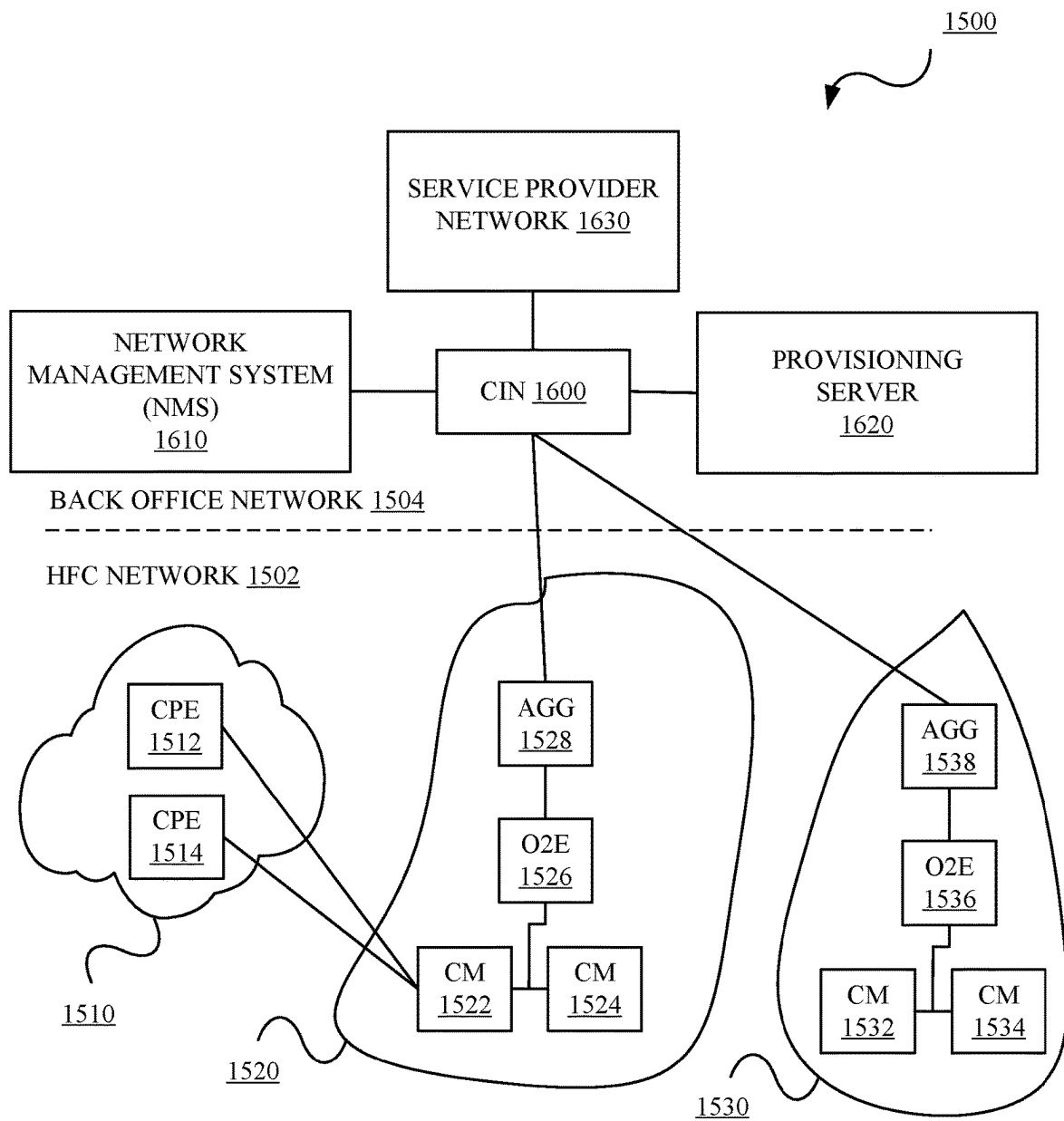


FIG. 1A

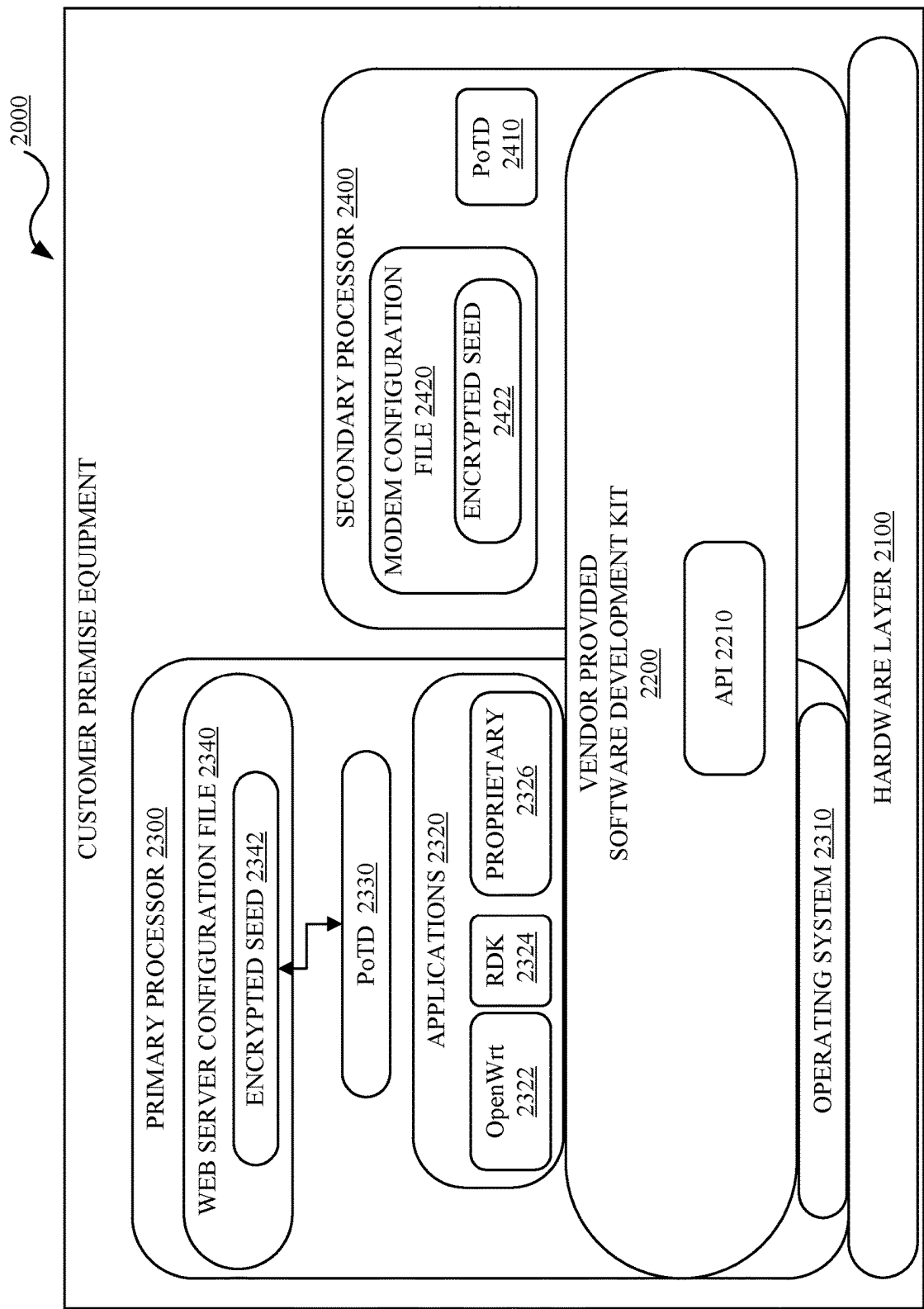


FIG. 2

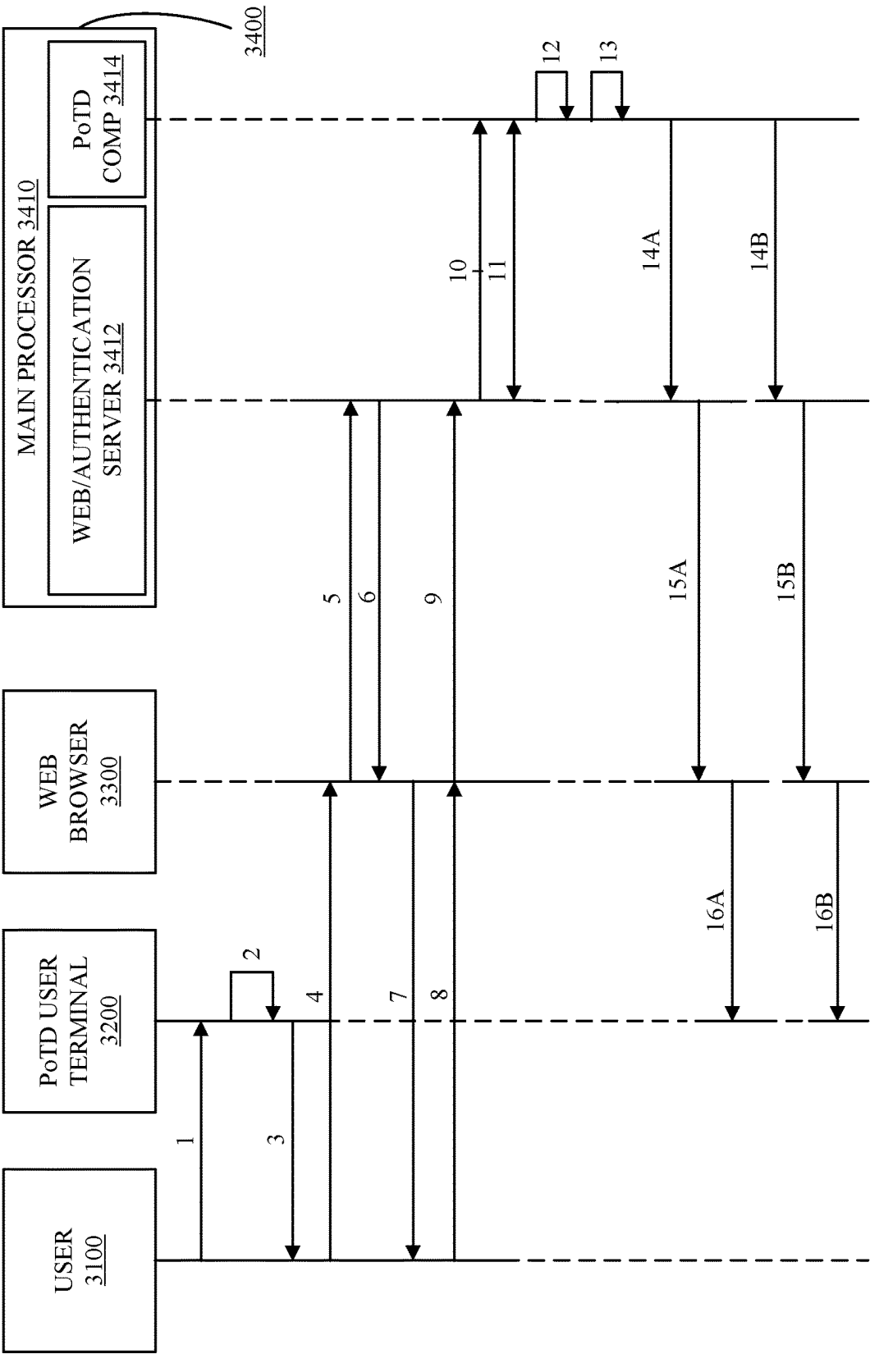


FIG. 3

3000

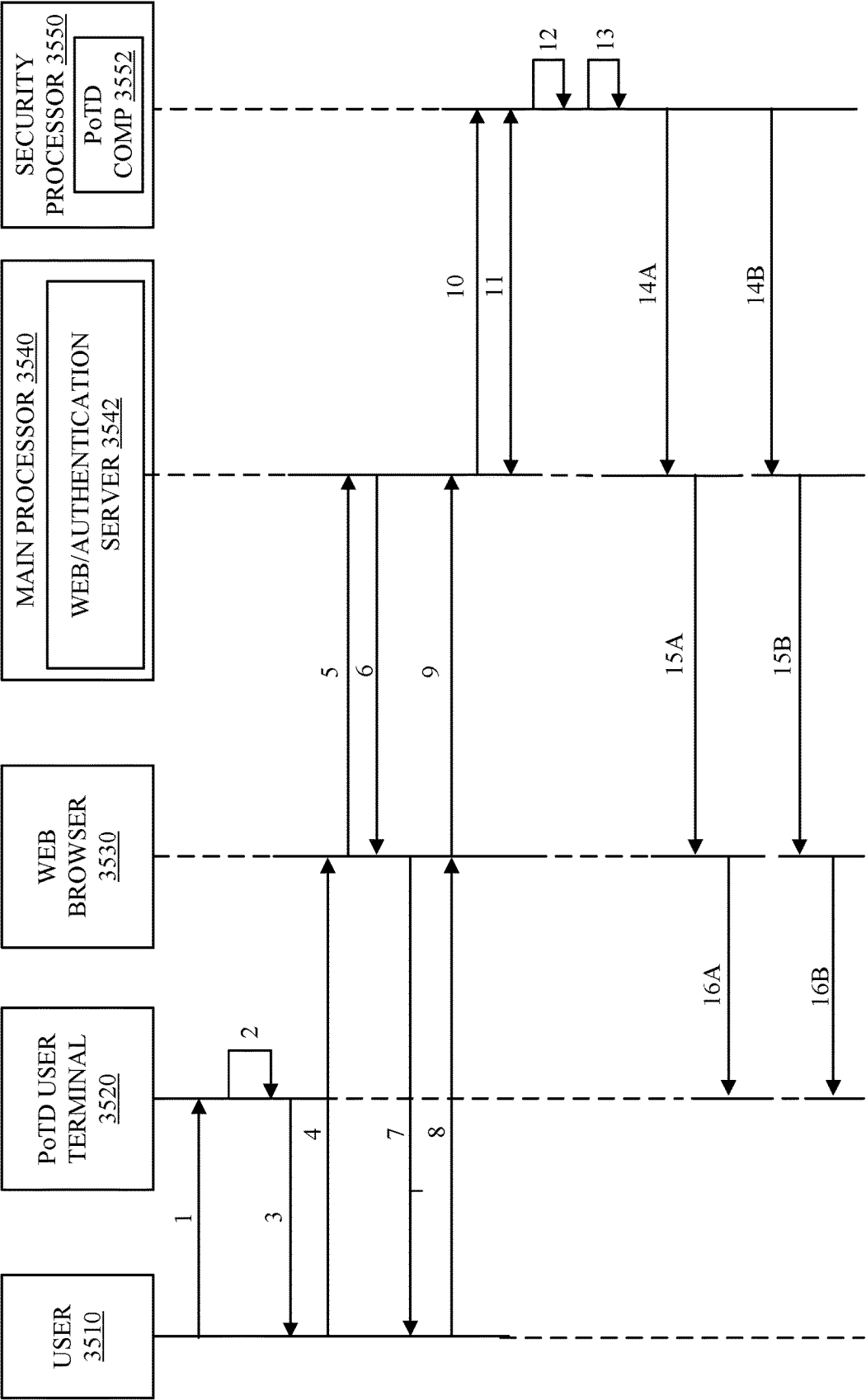


FIG. 3A

3500

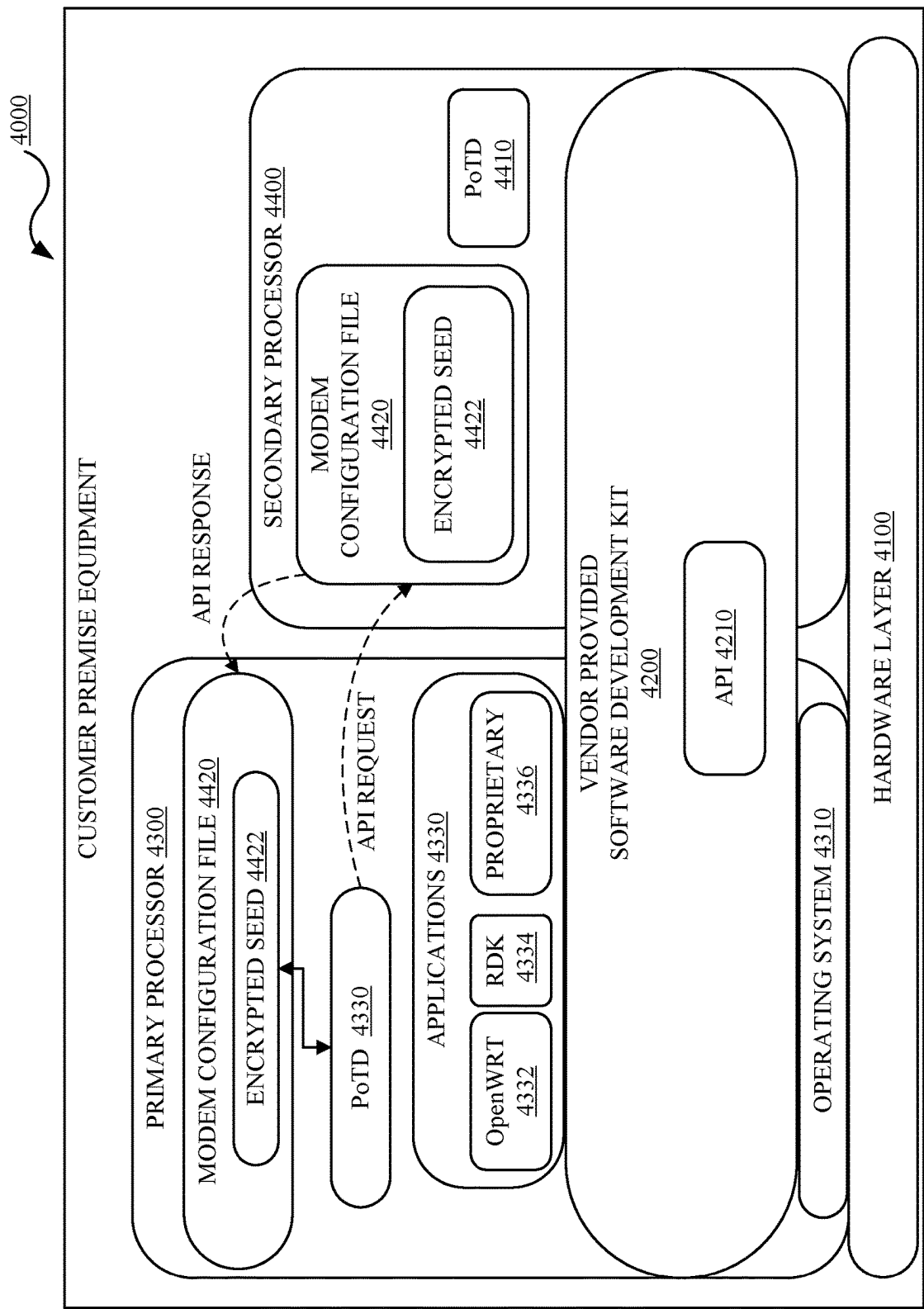


FIG. 4

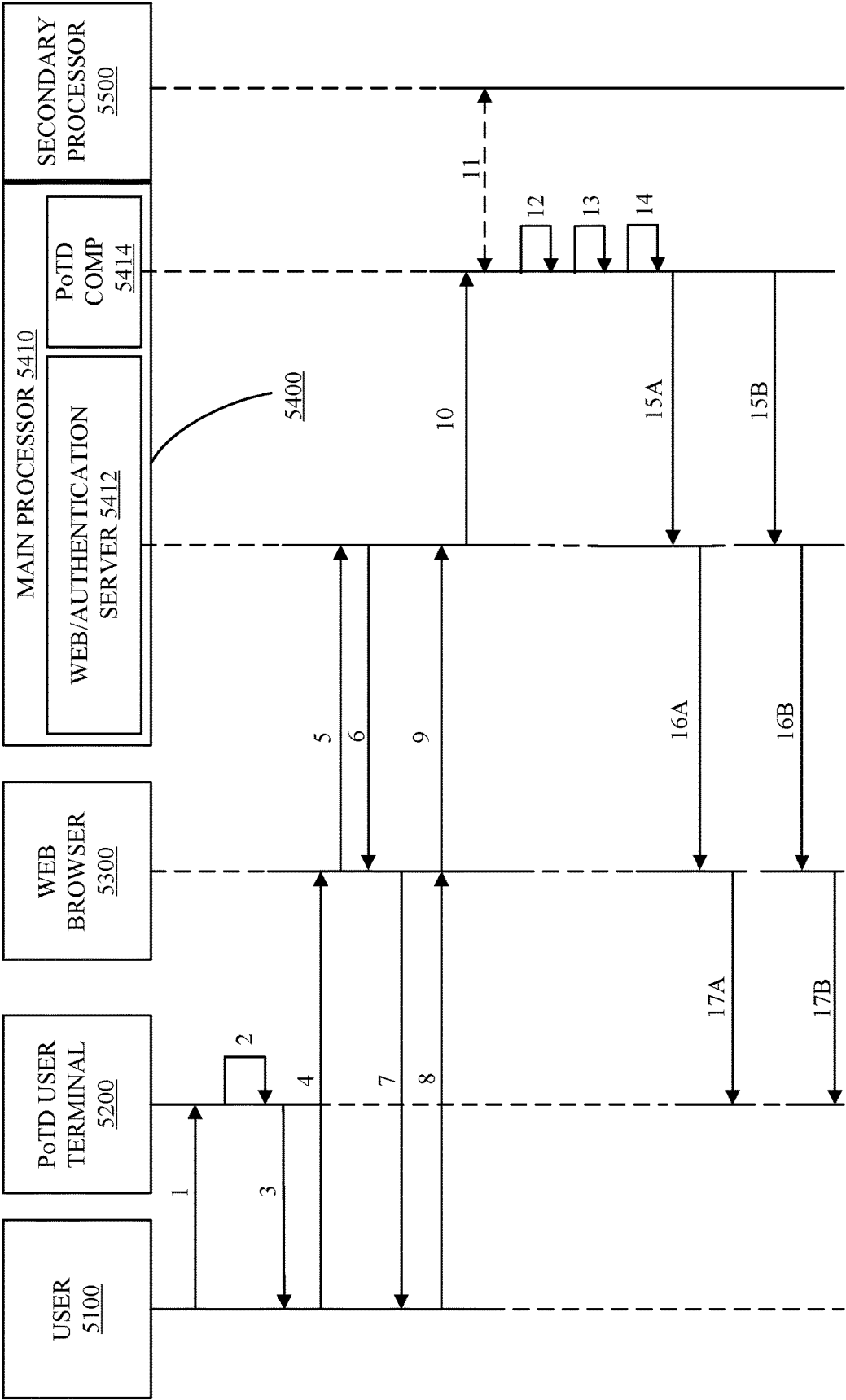


FIG. 5

5000

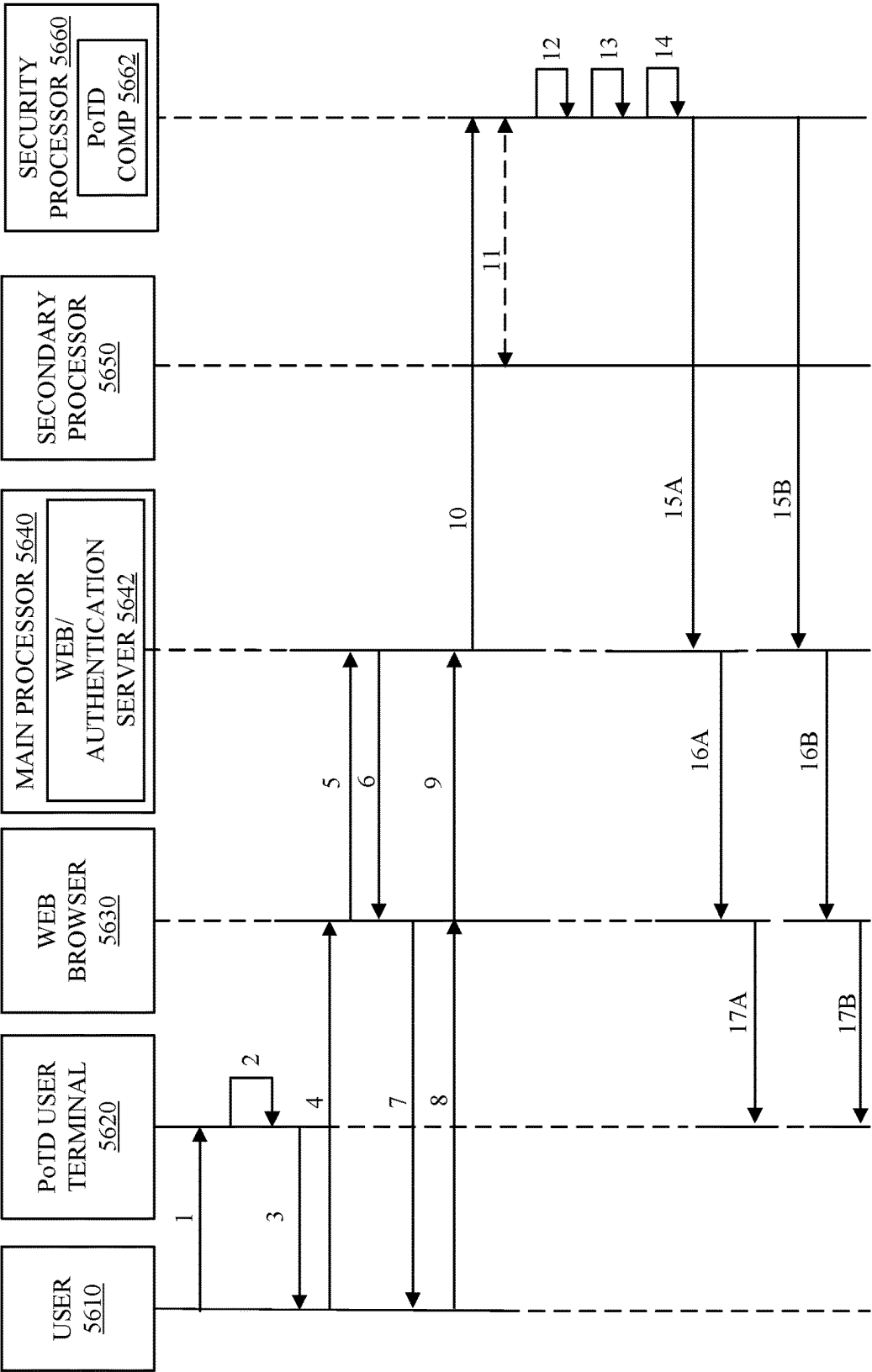


FIG. 5A

5600

6000

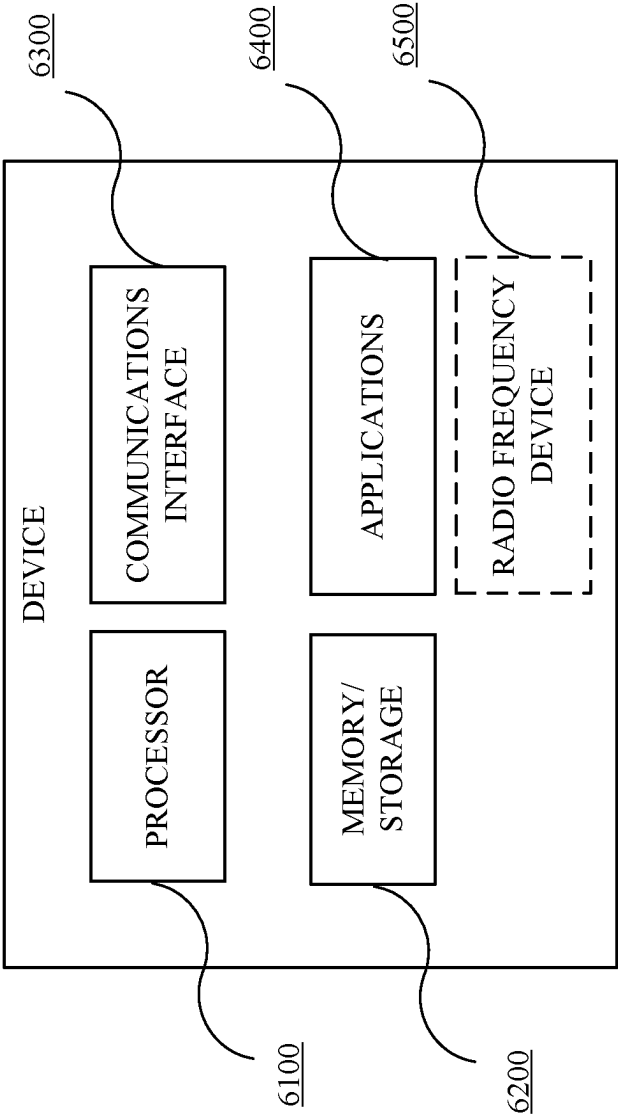


FIG. 6

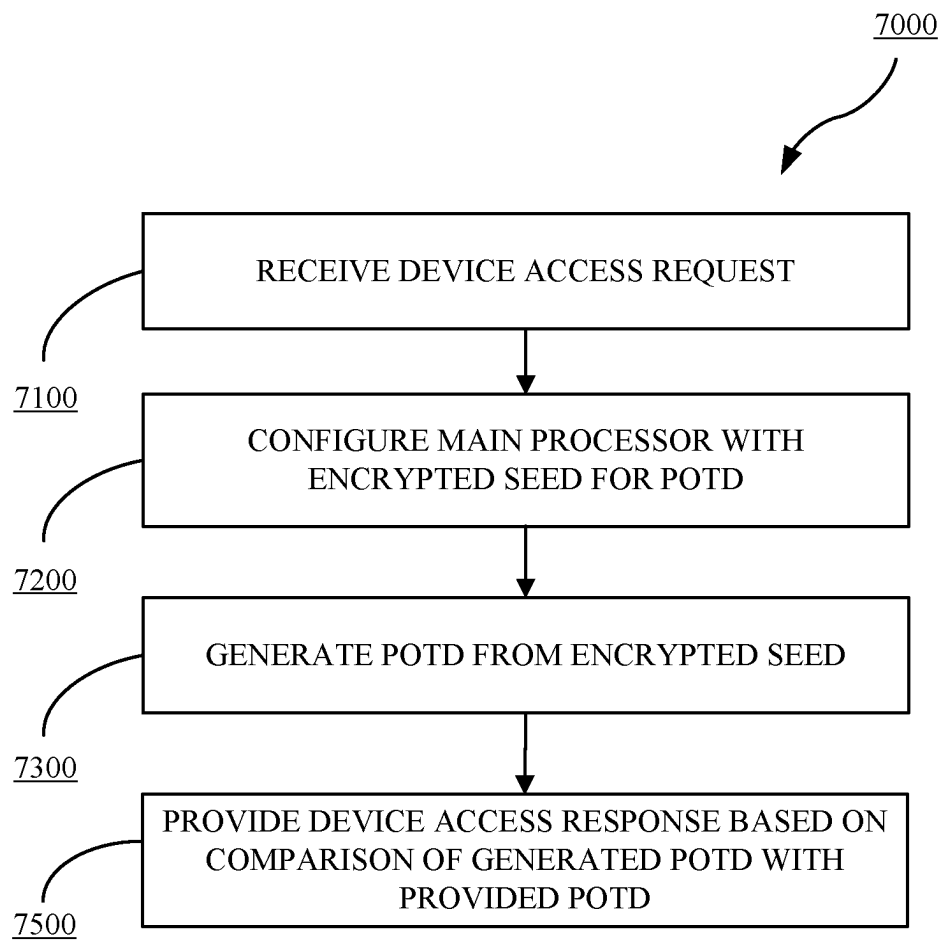


FIG. 7

1

CUSTOMER PREMISES EQUIPMENT ACCESS USING PASSWORD-OF-THE-DAY

TECHNICAL FIELD

This disclosure relates to network device access. More specifically, configuring a main processor in a customer premises equipment with an encrypted seed for password-of-the-day processing.

BACKGROUND

Service providers provide Internet and deliver content (collectively “services”) to its customers via multiple access customer premises equipment (CPE) which are connected to a service provider system via a coaxial cable system. The CPEs were developed by various silicon and original equipment manufacturer (OEM) vendors. These CPEs include monolithic and proprietary software stacks designed to meet a service providers’ specifications. One of the service providers’ specifications is security specifications which require the use of the Password-of-The-Day (PoTD). The PoTD, for example, enables a technician to remotely login and access the access CPE either via secure cryptographic network protocols such as secure shell (SSH) or Hypertext Transfer Protocol Secure (HTTPS) protocols.

Moving forward, service providers are developing their own software stacks including PoTD modules. Given that the current stacks are or include proprietary software stack components, solutions are needed which efficiently provide PoTD capabilities.

SUMMARY

Disclosed herein are methods and systems for configuring a main processor or security processor in a customer premises equipment with an encrypted seed for PoTD processing. In implementations, a customer premises equipment device includes a modem processor, a main processor in communication with the modem processor, and a password-of-the-day component included on the main processor. The main processor is configured to receive, via the modem processor from a user device, an access request to the customer premises equipment device. In response to the access request, the password-of-the-day component is configured to access an encrypted seed stored on the main processor, generate a password-of-the-day from the encrypted seed, and provide an access response based on comparison of the generated password-of-the-day with a password-of-the-day provided via the user device. A security processor can be used in lieu of the main processor. The PoTD component is then included in the security processor and the encrypted seed is then stored on the security processor.

BRIEF DESCRIPTION OF THE DRAWINGS

The disclosure is best understood from the following detailed description when read in conjunction with the accompanying drawings. It is emphasized that, according to common practice, the various features of the drawings are not to scale. On the contrary, the dimensions of the various features are arbitrarily expanded or reduced for clarity.

FIG. 1 is a diagram of an example of a network architecture in accordance with implementations of this disclosure.

2

FIG. 1A is a diagram of an example of a network architecture in accordance with implementations of this disclosure.

FIG. 2 is a block diagram of an example of a customer premises equipment in accordance with implementations of this disclosure.

FIG. 3 is a flow diagram of an example flow for configuring a main processor in a customer premises equipment with an encrypted seed for PoTD in accordance with implementations of this disclosure.

FIG. 3A is a flow diagram of an example flow for configuring a main processor in a customer premises equipment with an encrypted seed for PoTD processing with a security processor in accordance with implementations of this disclosure.

FIG. 4 is a block diagram of an example of a customer premises equipment in accordance with implementations of this disclosure.

FIG. 5 is a flow diagram of an example flow for configuring a main processor in a customer premises equipment with an encrypted seed for PoTD in accordance with implementations of this disclosure.

FIG. 5A is a flow diagram of an example flow for configuring a main processor in a customer premises equipment with an encrypted seed for PoTD processing with a security processor in accordance with implementations of this disclosure.

FIG. 6 is a block diagram of an example of a device in accordance with implementations of this disclosure.

FIG. 7 is a flowchart of an example method for configuring a main processor in a customer premises equipment with an encrypted seed for PoTD in accordance with implementations of this disclosure.

DETAILED DESCRIPTION

Reference will now be made in greater detail to implementations of the disclosure, examples of which are illustrated in the accompanying drawings. Wherever possible, the same reference numerals will be used throughout the drawings and the description to refer to the same or like parts.

As used herein, the terminology “server”, “computer”, “computing device or platform”, or “cloud computing system” includes any unit, or combination of units, capable of performing any method, or any portion or portions thereof, disclosed herein. For example, the “server”, “computer”, “computing device or platform”, or “cloud computing system” may include at least one or more processor(s).

As used herein, the terminology “processor” indicates one or more processors, such as one or more special purpose processors, one or more digital signal processors, one or more microprocessors, one or more controllers, one or more microcontrollers, one or more application processors, one or more central processing units (CPU)s, one or more graphics processing units (GPU)s, one or more digital signal processors (DSP)s, one or more application specific integrated circuits (ASIC)s, one or more application specific standard products, one or more field programmable gate arrays, any other type or combination of integrated circuits, one or more state machines, or any combination thereof.

As used herein, the terminology “memory” indicates any computer-usable or computer-readable medium or device that can tangibly contain, store, communicate, or transport any signal or information that may be used by or in connection with any processor. For example, a memory may be one or more read-only memories (ROM), one or more

random access memories (RAM), one or more registers, low power double data rate (LPDDR) memories, one or more cache memories, one or more semiconductor memory devices, one or more magnetic media, one or more optical media, one or more magneto-optical media, or any combination thereof.

As used herein, the terminology “instructions” may include directions or expressions for performing any method, or any portion or portions thereof, disclosed herein, and may be realized in hardware, software, or any combination thereof. For example, instructions may be implemented as information, such as a computer program, stored in memory that may be executed by a processor to perform any of the respective methods, algorithms, aspects, or combinations thereof, as described herein. For example, the memory can be non-transitory. Instructions, or a portion thereof, may be implemented as a special purpose processor, or circuitry, that may include specialized hardware for carrying out any of the methods, algorithms, aspects, or combinations thereof, as described herein. In some implementations, portions of the instructions may be distributed across multiple processors on a single device, on multiple devices, which may communicate directly or across a network such as a local area network, a wide area network, the Internet, or a combination thereof.

As used herein, the term “application” refers generally to a unit of executable software that implements or performs one or more functions, tasks, or activities. For example, applications may perform one or more functions including, but not limited to, telephony, web browsers, e-commerce transactions, media players, scheduling, management, smart home management, entertainment, and the like. The unit of executable software generally runs in a predetermined environment and/or a processor.

As used herein, the terminology “determine” and “identify,” or any variations thereof includes selecting, ascertaining, computing, looking up, receiving, determining, establishing, obtaining, or otherwise identifying or determining in any manner whatsoever using one or more of the devices and methods are shown and described herein.

As used herein, the terminology “example,” “the embodiment,” “implementation,” “aspect,” “feature,” or “element” indicates serving as an example, instance, or illustration. Unless expressly indicated, any example, embodiment, implementation, aspect, feature, or element is independent of each other example, embodiment, implementation, aspect, feature, or element and may be used in combination with any other example, embodiment, implementation, aspect, feature, or element.

As used herein, the terminology “or” is intended to mean an inclusive “or” rather than an exclusive “or.” That is, unless specified otherwise, or clear from context, “X includes A or B” is intended to indicate any of the natural inclusive permutations. That is, if X includes A; X includes B; or X includes both A and B, then “X includes A or B” is satisfied under any of the foregoing instances. In addition, the articles “a” and “an” as used in this application and the appended claims should generally be construed to mean “one or more” unless specified otherwise or clear from the context to be directed to a singular form.

Further, for simplicity of explanation, although the figures and descriptions herein may include sequences or series of steps or stages, elements of the methods disclosed herein may occur in various orders or concurrently. Additionally, elements of the methods disclosed herein may occur with other elements not explicitly presented and described herein. Furthermore, not all elements of the methods described

herein may be required to implement a method in accordance with this disclosure and claims. Although aspects, features, and elements are described herein in particular combinations, each aspect, feature, or element may be used independently or in various combinations with or without other aspects, features, and elements.

Further, the figures and descriptions provided herein may be simplified to illustrate aspects of the described implementations that are relevant for a clear understanding of the herein disclosed processes, machines, and/or manufactures, while eliminating for the purpose of clarity other aspects that may be found in typical similar devices, systems, and methods. Those of ordinary skill may thus recognize that other elements and/or steps may be desirable or necessary to implement the devices, systems, and methods described herein. However, because such elements and steps are well known in the art, and because they do not facilitate a better understanding of the disclosed implementations, a discussion of such elements and steps may not be provided herein. However, the present disclosure is deemed to inherently include all such elements, variations, and modifications to the described aspects that would be known to those of ordinary skill in the pertinent art in light of the discussion herein.

FIG. 1 is a diagram of an example network architecture 1000. The network architecture 1000 can include one or more customer premises equipment (CPE) 1100₁, 1100₂, . . . , 1100_N connected to or in communication with (collectively “connected to”) a service provider back-office system 1200 via a hybrid fiber coaxial cable (HFC), coaxial cable system, and/or combinations thereof (collectively “coaxial cable system”) 1300. The service provider back-office system 1200 can include service provider servers, networks, or clouds including, but not limited to, a provisioning server 1210, a network management system (NMS) 1220, and a Dynamic Host Configuration Protocol (DHCP) server 1230. The network architecture 1000 and the components therein may include other elements which may be desirable or necessary to implement the devices, systems, and methods described herein. However, because such elements and steps are well known in the art, and because they do not facilitate a better understanding of the disclosed implementations, a discussion of such elements and steps may not be provided herein.

The CPE 1100₁, 1100₂, . . . , 1100_N can be cable modems, Embedded Multimedia Terminal Adapters (eMTAs), optical network unit (ONU) devices, gateways, routers, set-top boxes, and the like which provides connectivity including Internet connectivity, wired connectivity, wireless connectivity, data, voice over IP, and combinations thereof. In implementations, the CPE 1100₁, 1100₂, . . . , 1100_N can include a voice gateway and external battery backup (EBBU) in case of external power failure. The CPE 1100₁, 1100₂, . . . , 1100_N can be deployed, for example, at a customer premises, residence, office, and the like.

The service provider back-office system 1200 can include multiple components to provide services to customers via the CPE 1100₁, 1100₂, . . . , 1100_N.

The provisioning server 1210 can provide configuration information and data to components in the network architecture 1000 including, for example, the CPE 1100₁, 1100₂, . . . , 1100_N. The configuration information and data enable operation of the CPE 1100₁, 1100₂, . . . , 1100_N. The NMS 1220 can include applications which monitor, maintain, and optimize a network. The DHCP server 1230 can manage Internet Protocol (IP) addresses it allocates to network nodes.

5

FIG. 1A is a diagram of an example network architecture **1500**. The network architecture **1500** can include one or more CPEs **1512** and **1514** deployed on a local area network (LAN) **1510** and connected to network components **1520** in a hybrid fiber-coaxial (HFC) network **1502**. The HFC network **1502** can include any number of network components including, but not limited to, network components **1520** and **1530**. The network components **1520** and **1530** can be connected to a converged interconnect network (CIN) **1600** in a service provider's back-office network **1504**. The service provider's back-office network **1504** can include, but is not limited to, an NMS **1610**, a provisioning system **1620**, and a service provider cloud network **1630**. The network architecture **1500** and the components therein may include other elements which may be desirable or necessary to implement the devices, systems, and methods described herein. However, because such elements and steps are well known in the art, and because they do not facilitate a better understanding of the disclosed implementations, a discussion of such elements and steps may not be provided herein.

The CPEs **1512** and **1514** can be cable modems, Embedded Multimedia Terminal Adapters (eMTAs), optical network unit (ONU) devices, gateways, routers, set-top boxes, and the like which provides connectivity including Internet connectivity, wired connectivity, wireless connectivity, data, voice over IP, and combinations thereof. In implementations, the CPEs **1512** and **1514** can include a voice gateway and external battery backup (EBBU) in case of external power failure. The CPEs **1512** and **1514** can be deployed, for example, at a customer premises, residence, office, and the like.

The provisioning server **1610** can provide configuration information and data to components in the network architecture **1500** including, for example, the CPEs **1512** and **1514**. The configuration information and data enable operation of the CPEs **1512** and **1514**. The NMS **1610** can include applications which monitor, maintain, and optimize a network.

The network components **1520** and **1530** can include, but are not limited to, cable modems (CM) **1522**, **1524**, **1532**, and **1534**, optical-to-electrical (O2E) converters **1526** and **1536**, and aggregators **1528** and **1538**, respectively. The aggregators **1528** and **1538** can be, for example, a CMTS or a Converged Cable Access Platform (CCAP).

FIG. 2 is a block diagram of an example CPE **2000** in accordance with implementations of this disclosure. In implementations, the CPE **2000** can be the CPE **1100**₁, **1100**₂, . . . , **1100**_N of FIG. 1. The CPE **2000** implements a hardware layer **2100**, a vendor provided software development kit (SDK) **2200**, and a dual processor configuration including a router/gateway, main, or primary processor (collectively "gateway processor") **2300** and a cable modem or secondary processor (collectively "cable modem processor") **2400**.

The gateway processor **2300** includes a stack which includes, but is not limited to, an operating system **2310**, applications **2320**, a PoTD module, component, generator, or software (collectively "PoTD component") **2330**, and a web server configuration file **2340**. The applications **2320** can include, but is not limited to, an OpenWrt Linux operating system **2322**, a Reference Design Kit (RDK) **2324**, and proprietary software **2326**. The web server configuration file **2340** includes an encrypted seed **2342**. The gateway processor **2300** is provisioned with the web server configuration file **2340** during registration and provisioning by a provisioning server, such as the provisioning server

6

1210. The gateway processor **2300** is directed to the operation and functionality of the CPE **2000**.

The cable modem processor **2400** includes a stack which includes, but is not limited to, a PoTD component **2410** and a modem configuration file **2420**. The modem configuration file **2420** can include, but is not limited to, configuration information and an encrypted seed **2422**. The cable modem processor **2400** is provisioned with the modem configuration file **2420** during registration and provisioning by a provisioning server, such as the provisioning server **1210**. The cable modem processor **2400** is directed to the operation and functionality of Data Over Cable Service Interface Specification (DOCSIS) protocol suites including, but not limited to, DOCSIS 4.0 and DOCSIS 3.1.

The gateway processor **2300** is a more powerful processor than the cable modem processor **2400**. That is, the cable modem processor **2400** is limited in capability and functionality.

The SDK **2200** can include, but is not limited to, Application Programming Interfaces (APIs) **2210** which provide communication connectivity between various components in the CPE **2000** including between the hardware layer **2100**, the gateway processor **2300**, and/or the cable modem processor **2400**. For example, the SDK **2220** can include proprietary APIs to enable an end-user to access limited DOCSIS information available from the cable modem processor **2400** to the gateway processor **2300**, provide secure boot-up of the silicon, check the device flash/DRAM memories and memory partitions, load the device tree, and establish a Chain-of-Trust (CoT) between the silicon and the applications.

A user or technician can access the CPE **2000** using a user terminal, laptop, or device (collectively "user device") which also includes a PoTD component. The CPE **2000** can be accessed via SSH or HTTPS protocols using the PoTD generated by the PoTD component on the user device.

When accessing the CPE **2000** using the SSH protocol, the user uses the PoTD component on the user device to generate a PoTD, password, or credential (collectively "PoTD"). The user device transmits or sends a SSH protocol request, which is received and processed by the cable modem processor **2400**. The user sends the PoTD via the user device responsive to the cable modem processor **2400**. The PoTD component **2410** fetches the encrypted seed **2422** from the modem configuration file **2420** to generate a PoTD in response to the request. The PoTD component **2410** then compares, verifies, and/or authenticates (collectively "verifies" and referred to as verification processing) the generated PoTD against the user provided PoTD to determine if the user is granted or denied access to the CPE **2000**. If the entered password matches a generated password, the PoTD component **2410** allows the user to access the CPE **2000**. If the entered password does not match the generated password, the PoTD component **2410** denies the user access to the CPE **2000**. The PoTD processing is performed in the cable modem processor **2400**.

When accessing the CPE **2000** using the HTTPS protocol, the user uses the PoTD component on the user device to generate a PoTD. The user uses a web browser on the user device to transmits or sends an authentication request, which is received and processed by the cable modem processor **2400**. Since this is a web based or browser based protocol, the cable modem processor **2400** passes the authentication request to the gateway processor **2300**. That is, the web server runs on the gateway processor **2300** and not on the cable modem processor **2400**. The gateway processor **2300** has to execute one or more new or additional remote

7

procedure calls (RPCs) to the PoTD component **2410** to perform the verification processing. This is because the PoTD component **2410** has access to the encrypted seed **2422**. The gateway processor **2300** does not have access to an encrypted seed. The gateway processor **2300** cannot perform the verification process without the assistance of the cable modem processor **2400** and the new or additional RPCs.

To eliminate the new or additional RPCs, the encrypted seed **2342** can be included in the web server configuration file **2340** during the firmware image build of the gateway processor **2300**. That is, the web server configuration file **2340** includes an encrypted seed **2342** which is built into the firmware of the gateway processor **2300**. The gateway processor **2300** is provisioned by a provisioning server, such as the provisioning server **1210**. As a result, the gateway processor **2300** and the PoTD component **2330** can perform the verification processing without the need of the new or additional RPCs. A new image or code based would need to be downloaded to the CPE **2000** to update the encrypted seed **2342**.

FIG. 3 is a flow diagram of an example flow **3000** for configuring a main processor in a customer premises equipment with an encrypted seed for PoTD in accordance with implementations of this disclosure. The flow **3000** is operable between a user **3100**, a PoTD component **3200** on a user terminal or device, a web browser **3300**, and a CPE **3400** including a main processor **3410** which has a web server **3412** and a PoTD component **3414**. The CPE **3400** can be the CPE **2000** of FIG. 2 and the CPE **1100₁**, **1100₂**, . . . , **1100_N** as used in FIG. 1.

The user **3100** requiring access to the CPE **3400** can execute the PoTD component **3200** (1). The PoTD component **3200** can generate a PoTD based on an encrypted seed in the PoTD component **3200** (which is the same value stored on the CPE **3400**) and a current date (2). The generated PoTD can be displayed or made visible to the user **3100** (3). The user **3100** can open or access the appropriate webpage of the CPE **3400** via the web browser **3300** (4), which in turn can send an authentication and/or webpage request to the web server **3412** (5). The web server **3412** can perform a redirect message to the web browser **3300** (6) to request a credential form from the user **3100** (7). The user **3100** can enter the PoTD into the web browser (8), which in turn can forward the PoTD to the web server **3412** for authentication or verification (9). The web server **3412** can forward the PoTD to the PoTD component **3414** (10), which in turn can fetch an encrypted seed from the web configuration file (11). The PoTD component **3414** can generate the PoTD using the encrypted seed and the current date (12). The PoTD module **3414** can compare the generated PoTD with the user provided PoTD (13). In the event that the generated PoTD matches the user provided PoTD (14A), the web server **3412** can send an authorization message to the web browser **3300** (15A), which in turn can grant the user **3100** access to the CPE **3400** (16A). In implementations, the CPE **3400** can include a timer to limit an access session duration to a defined period of time. In implementations, the defined period of time can range from 15 to 60 minutes. In implementations, the defined period of time is dynamic, configurable, and/or combinations thereof. In the event that the generated PoTD does not match the user provided PoTD (14B), the web server **3412** can send a denial message to the web browser **3300** (15B), which in turn can deny the user **3100** access to the CPE **3400** (16B). In implementations, the CPE **3400** can include a counter to limit or set the number of retries a user has to gain access before the user is locked

8

out. In implementations, the counter is dynamic, configurable, and/or combinations thereof.

FIG. 3A is a flow diagram of an example flow **3500** for CPE processing with an encrypted seed for PoTD processing using a security processor in accordance with implementations of this disclosure. The flow **3500** is operable between a user **3510**, a PoTD component **3520** on a user terminal or device, a web browser **3530**, and a CPE including a main processor **3540** which has a web server **3542** and security processor **3550** which has a PoTD component **3552**. The CPE can be the CPE **2000** of FIG. 2 and the CPE **1100₁**, **1100₂**, . . . , **1100_N** as used in FIG. 1, with the inclusion of the security processor **3550** in a multi-processor configuration.

The user **3510** requiring access to the CPE can execute the PoTD component **3520** (1). The PoTD component **3520** can generate a PoTD based on an encrypted seed in the PoTD component **3520** (which is the same value stored on the CPE) and a current date (2). The generated PoTD can be displayed or made visible to the user **3510** (3). The user **3510** can open or access the appropriate webpage of the CPE via the web browser **3530** (4), which in turn can send an authentication and/or webpage request to the web server **3542** (5). The web server **3542** can perform a redirect message to the web browser **3530** (6) to request a credential form from the user **3510** (7). The user **3510** can enter the PoTD into the web browser **3530** (8), which in turn can forward the PoTD to the web server **3542** for authentication or verification (9). The web server **3542** can forward the PoTD to the PoTD component **3552** (10), which in turn can fetch an encrypted seed from the web configuration file on the main processor **3542** (11). The PoTD component **3552** can generate the PoTD using the encrypted seed and the current date (12). The PoTD module **3552** can compare the generated PoTD with the user provided PoTD (13). In the event that the generated PoTD matches the user provided PoTD (14A), the web server **3542** can send an authorization message to the web browser **3530** (15A), which in turn can grant the user **3510** access to the CPE (16A). In implementations, the CPE can include a timer to limit an access session duration to a defined period of time. In implementations, the defined period of time can range from 15 to 60 minutes. In implementations, the defined period of time is dynamic, configurable, and/or combinations thereof. In the event that the generated PoTD does not match the user provided PoTD (14B), the web server **3542** can send a denial message to the web browser **3530** (15B), which in turn can deny the user **3510** access to the CPE (16B). In implementations, the CPE can include a counter to limit or set the number of retries a user has to gain access before the user is locked out. In implementations, the counter is dynamic, configurable, and/or combinations thereof.

FIG. 4 is a block diagram of an example of a CPE **4000** in accordance with implementations of this disclosure. In implementations, the CPE **4000** can be the CPE **1100₁**, **1100₂**, . . . , **1100_N** of FIG. 1. The CPE **4000** implements a hardware layer **4100**, a vendor provided SDK **4200**, and a dual processor configuration including a router/gateway, main, or primary processor (collectively "gateway processor") **4300** and a cable modem or secondary processor (collectively "cable modem processor") **4400**.

The gateway processor **4300** includes a stack which includes, but is not limited to, an operating system **4310**, applications **4320**, and a PoTD module, component, generator, or software (collectively "PoTD component") **4330**. The applications **4320** can include, but is not limited to, an OpenWrt Linux operating system **4332**, a RDK **4334**, and

proprietary software **4336**. The gateway processor **4300** is directed to the operation and functionality of the CPE **4000**.

The cable modem processor **4400** includes a stack which includes, but is not limited to, a PoTD component **4410** and a modem configuration file **4420**. The modem configuration file **4420** can include, but is not limited to, configuration information and an encrypted seed **4422**. The cable modem processor **4400** is provisioned with the modem configuration file **4420** during registration and provisioning by a provisioning server, such as the provisioning server **1210**. The cable modem processor **4400** is directed to the operation and functionality of DOCSIS protocol suites including, but not limited to, PacketCable 1.5/2.0.

The gateway processor **4300** is a more powerful processor than the cable modem processor **4400**. That is, the cable modem processor **4400** is limited in capability and functionality.

The SDK **4200** can include, but is not limited to, APIs **4210** which provide communication connectivity between various components in the CPE **4000** including between the hardware layer **4100**, the gateway processor **4300**, and/or the cable modem processor **4400**. For example, the SDK **4200** can include proprietary APIs to enable an end-user to access limited DOCSIS information available from the cable modem processor **2400** to the gateway processor **2300**.

As described with respect to FIG. 2, a user or technician can access the CPE **4000** using a user terminal, laptop, or device (collectively "user device") which also includes a PoTD component. The CPE **4000** can be accessed via SSH as described with respect to FIG. 2.

To eliminate the new or additional RPCs when accessing via HTTPS as described with respect to FIG. 2, the gateway processor **4300** can determine if the modem configuration file **4420** is already available and stored in the gateway processor **4300** as a result of other processing requiring the configuration information in the modem configuration file **4420**. If the modem configuration file **4420** is not stored in the gateway processor **4300**, the gateway processor **4300** can use the already provided API **4210** to request the modem configuration file **4420**. The API request is a one time RPC. In implementations, updates to the encrypted seed **4422** can be pushed out to the CPE **4000** by the service provider back-office system **1200** and/or the provisioning server **1210**, as appropriate and as needed. The modem configuration file **4420** is then stored in the gateway processor **4300**. The PoTD component **4330** can extract the encrypted seed **4422** from the modem configuration file **4420** and store as a global variable in or on the gateway processor **4300**. The PoTD component **4330** can then use the encrypted seed **4422** to generate a PoTD as needed. In both instances, as a result, the gateway processor **4300** and the PoTD component **4330** can perform the verification processing without the need of the new or additional RPCs.

FIG. 5 is a flow diagram of an example flow **5000** for configuring a main processor in a customer premises equipment with an encrypted seed for PoTD in accordance with implementations of this disclosure. The flow **5000** is operable between a user **5100**, a PoTD component **5200** on a user terminal or device, a web browser **5300**, a CPE **5400** including a main processor **5410** which has a web server **5412** and a PoTD component **5414**, and a secondary or cable processor **5500**. The CPE **5400** can be the CPE **4000** of FIG. 4 and the CPE **1100₁**, **1100₂**, . . . , **1100_N** as used in FIG. 1.

The user **5100** requiring access to the CPE **5400** can execute the PoTD component **5200** (1). The PoTD component **5200** can generate a PoTD based on an encrypted seed in the PoTD component **5200** (which is the same value

stored on the CPE **5400**) and a current date (2). The generated PoTD can be displayed or made visible to the user **5100** (3). The user **5100** can open or access the appropriate webpage of the CPE **5400** via the web browser **5300** (4), which in turn can send an authentication and/or webpage request to the web server **5412** (5). The web server **5412** can perform a redirect message to the web browser **5300** (6) to request a credential form from the user **5100** (7). The user **5100** can enter the PoTD into the web browser (8), which in turn can forward the PoTD to the web server **5412** for authentication or verification (9). The web server **5412** can forward the PoTD to the PoTD component **5414** (10). The PoTD component **5414** can determine if the modem configuration file **4420** is already available on the main processor **5410**, extract an encrypted seed if the modem configuration file **4420** is available, and store as a global variable. This can be a one-time operation. If the modem configuration file **4420** is not available on the main processor **5410**, then the PoTD component **5414** can make an API call to the secondary or cable processor **5500** to get the modem configuration file (11). The PoTD component **5414** can then fetch the encrypted seed from the modem configuration file and store the encrypted seed as a global variable in the PoTD component **5414** on the main processor **5410** (12). In implementations, the PoTD component **5414** can delete the modem configuration file on the main processor **5410**. The PoTD component **5414** can generate the PoTD using the encrypted seed and the current date (13). The PoTD component **5414** can compare the generated PoTD with the user provided PoTD (14). In the event that the generated PoTD matches the user provided PoTD (15A), the web server **5412** can send an authorization message to the web browser **5300** (16A), which in turn can grant the user **5100** access to the CPE **5400** (17A). In implementations, the CPE **5400** can include a timer to limit an access session duration to a defined period of time. In implementations, the defined period of time can range from 15 to 60 minutes. In implementations, the defined period of time is dynamic, configurable, and/or combinations thereof. In the event that the generated PoTD does not match the user provided PoTD (15B), the web server **5412** can send a denial message to the web browser **5300** (16B), which in turn can deny the user **5100** access to the CPE **5400** (17B). In implementations, the CPE **5400** can include a counter to limit or set the number of retries a user has to gain access before the user is locked out. In implementations, the counter is dynamic, configurable, and/or combinations thereof.

The flow **5000** can use an already available API call (RPC call) to the secondary processor to obtain the modem configuration file. The flow **5000** does not need to create a new RPC to pass and get verified user entered password. The flow **5000** can enable the PoTD component **5414** to use a dynamic encrypted seed from the modem configuration file if and when the modem configuration file is updated.

FIG. 5A is a flow diagram of an example flow **5600** for CPE processing an encrypted seed for PoTD processing via a security processor in accordance with implementations of this disclosure. The flow **5600** is operable between a user **5610**, a PoTD component **5620** on a user terminal or device, a web browser **5630**, a CPE including a main processor **5640** which has a web server **5642**, a secondary or cable processor **5500**, and a security processor **5660** which has a PoTD component **5662**. The CPE can be the CPE **4000** of FIG. 4 and the CPE **1100₁**, **1100₂**, . . . , **1100_N** as used in FIG. 1, with the inclusion of the security processor **3550** in a multi-processor configuration.

11

The user **5610** requiring access to the CPE can execute the PoTD component **5620** (1). The PoTD component **5620** can generate a PoTD based on an encrypted seed in the PoTD component **5620** (which is the same value stored on the CPE) and a current date (2). The generated PoTD can be displayed or made visible to the user **5610** (3). The user **5610** can open or access the appropriate webpage of the CPE via the web browser **5630** (4), which in turn can send an authentication and/or webpage request to the web server **5642** (5). The web server **5642** can perform a redirect message to the web browser **5630** (6) to request a credential form from the user **5610** (7). The user **5610** can enter the PoTD into the web browser **5630** (8), which in turn can forward the PoTD to the web server **5642** for authentication or verification (9). The web server **5642** can forward the PoTD to the PoTD component **5662** (10). The PoTD component **5662** can determine if the modem configuration file **4420** is already available on the main processor **5410** and/or security processor **5660**, extract an encrypted seed if the modem configuration file **4420** is available, and store as a global variable. The encrypted seed can be stored locally on the security processor **5660**. That is, the encrypted seed can be stored locally where the PoTD component is located. This can be a one-time operation. In the implementations described herein, if the CPE is reprovisioned, then this operation is done again. If the modem configuration file **4420** is not available on the main processor **5640** and/or security processor **5660**, then the PoTD component **5662** can make an API call to the secondary processor **5650** to get the modem configuration file (11). The PoTD component **5662** can then fetch the encrypted seed from the modem configuration file and store the encrypted seed as a global variable in the PoTD component **5662** (12). As described, the encrypted seed can be stored locally on the security processor **5660**. That is, the encrypted seed can be stored locally where the PoTD component is located. In implementations, the PoTD component **5662** can delete the modem configuration file once the encrypted seed is obtained. The PoTD component **5662** can generate the PoTD using the encrypted seed and the current date (13). The PoTD component **5662** can compare the generated PoTD with the user provided PoTD (14). In the event that the generated PoTD matches the user provided PoTD (15A), the web server **5642** can send an authorization message to the web browser **5630** (16A), which in turn can grant the user **5610** access to the CPE (17A). In implementations, the CPE can include a timer to limit an access session duration to a defined period of time. In implementations, the defined period of time can range from 15 to 60 minutes. In implementations, the defined period of time is dynamic, configurable, and/or combinations thereof. In the event that the generated PoTD does not match the user provided PoTD (15B), the web server **5642** can send a denial message to the web browser **5630** (16B), which in turn can deny the user **5610** access to the CPE (17B). In implementations, the CPE can include a counter to limit or set the number of retries a user has to gain access before the user is locked out. In implementations, the counter is dynamic, configurable, and/or combinations thereof.

The flow **5600** can use an already available API call (RPC call) to the secondary processor **5650** to obtain the modem configuration file. The flow **5600** does not need to create a new RPC to pass and get verified user entered password. The flow **5600** can enable the PoTD component **5662** to use a dynamic encrypted seed from the modem configuration file if and when the modem configuration file is updated.

12

FIG. 6 is a block diagram of an example of a device **6000** in accordance with implementations of this disclosure. The device **6000** may include, but is not limited to, a processor **6100**, a memory/storage **6200**, a communication interface **6300**, and applications **6400**. In an implementation, the device **6000** may include a radio frequency device **6500**. The device **6000** may include or implement, for example, the CPE **1100**₁, **1100**₂, . . . , **1100**_N, the service provider back-office system **1200**, the provisioning server **1210**, the components in the network architecture **1500**, the CPE **2000**, the PoTD component **3200** on the user terminal or device, the web browser **3300**, the CPE **3400**, the components in the flow **3500**, the CPE **4000**, the PoTD component **5200** on the user terminal or device, the web browser **5300**, the CPE **5400**, and the components in the flow **5600**. The applicable or appropriate techniques or methods described herein may be stored in the memory/storage **6200** and executed by the processor **6100** in cooperation with the memory/storage **6200**, the communications interface **6300**, the applications **6400**, and the radio frequency device **6500** (when applicable), as appropriate. The device **6000** may include other elements which may be desirable or necessary to implement the devices, systems, and methods described herein. However, because such elements and steps are well known in the art, and because they do not facilitate a better understanding of the disclosed implementations, a discussion of such elements and steps may not be provided herein.

FIG. 7 is a flowchart of an example method **7000** for configuring a main processor in a CPE with an encrypted seed for PoTD processing in accordance with implementations of this disclosure. The method **7000** includes: receiving **7100** a device access request; configuring **7200** the main processor with an encrypted seed for PoTD processing; generating **7300** a PoTD from the encrypted seed; and providing **7400** access response based on comparison of generated PoTD with provided PoTD. The method **7000** can be implemented, for example, in the CPE **1100**₁, **1100**₂, . . . , **1100**_N, the service provider back-office system **1200**, the provisioning server **1210**, the components in the network architecture **1500**, the CPE **2000**, the PoTD component **3200** on the user terminal or device, the web browser **3300**, the CPE **3400**, the components in the flow **3500**, the CPE **4000**, the PoTD component **5200** on the user terminal or device, the web browser **5300**, the CPE **5400**, the components in the flow **5600**, the device **6000**, the processor **6100**, the memory/storage **6200**, the communications interface **6300**, the applications **6400**, and the radio frequency device **6500** when available, as appropriate and applicable.

The method includes receiving **7100** a device access request. A user using a user device can send an access request to an access device such as a CPE. The user device includes a PoTD component which includes or has access to an encrypted seed used by the CPE. The PoTD component can generate a PoTD. The user can send the PoTD as requested during the verification process.

The method includes configuring **7200** a main processor in access device with an encrypted seed for PoTD processing. The device access includes a dual processor configuration with a main processor and a modem processor. The main processor has more capabilities than the modem processor in terms of processor power, computational power, memory, and other processing features. The modem processor handles data interface communications related to DOCSIS and other protocols. The main processor handles all other processing. For web based device access, a PoTD component on the main processor needs access to an encrypted seed. In implementations, the encrypted seed can

13

be provided in a web configuration file in the main processor. In implementations, the PoTD component on the main processor can check if a modem configuration file is present on the main processor. If the modem configuration file is available, the PoTD component can fetch the encrypted seed from the modem configuration file. If the modem configuration file is not available, the main processor and/or the PoTD component can run an API call to obtain and store the modem configuration file on the main processor. The PoTD component on the main processor can then fetch and store the encrypted seed as a global variable for PoTD generation on the main processor. In implementations, a security processor can be used for PoTD processing as described herein. That is, the PoTD component is on the security processor instead of being on the main or primary processor. The encrypted seed can then also be stored on the security processor. That is, in some implementations, the encrypted seed is stored local to where the PoTD component is located.

The method includes generating **7300** a PoTD from the encrypted seed. The PoTD component on the main processor can generate a PoTD using the encrypted seed based on the received access request.

The method includes providing **7400** device access response based on comparison of generated PoTD with provided PoTD. The PoTD component on the main processor can compare the generated PoTD with a user provided PoTD. If the generated PoTD and the user provided PoTD match, the PoTD component on the main processor grants access. If the generated PoTD and the user provided PoTD does not match, the PoTD component on the main processor denies access. In implementations, a CPE can include a timer to limit an access session duration to a defined period of time. In implementations, the defined period of time can range from 15 to 60 minutes. In implementations, the defined period of time is dynamic, configurable, and/or combinations thereof. In implementations, the CPE can include a counter to limit or set the number of retries a user has to gain access before the user is locked out. In implementations, the counter is dynamic, configurable, and/or combinations thereof.

Disclosed herein are methods and systems for configuring a main processor or security processor in a customer premises equipment with an encrypted seed for PoTD processing. In implementations, a customer premises equipment device includes a modem processor, a main processor in communication with the modem processor, and a password-of-the-day component included on a security processor when present or the main processor. The main processor is configured to receive, via the modem processor from a user device, an access request to the customer premises equipment device. In response to the access request, the password-of-the-day component is configured to access an encrypted seed stored on the main processor when the password-of-the-day component is included on the main processor or access the encrypted seed stored on the security processor when the password-of-the-day component is included on the security processor when present, generate a password-of-the-day from the encrypted seed, and provide an access response based on comparison of the generated password-of-the-day with a password-of-the-day provided via the user device.

In implementations, a web server configuration file on the main processor is provisioned with the encrypted seed and the password-of-the-day component is configured to fetch the encrypted seed from the web server configuration file. In implementations, the password-of-the-day component is further configured to fetch the encrypted seed from a modem

14

configuration file previously obtained from the modem processor by the main processor or the security processor when present, and store the encrypted seed as a global variable for access by the password-of-the-day component for password-of-the-day generation. In implementations, the password-of-the-day component is further configured to get a modem configuration file from the modem processor to the main processor or the security processor when present using an available Application Programming Interface call, fetch the encrypted seed from the modem configuration file on the main processor or the security processor when present, and store the encrypted seed as a global variable for access by the password-of-the-day component for password-of-the-day generation. In implementations, the modem configuration file is obtained once for a new or updated encrypted seed. In implementations, the password-of-the-day component is further configured to delete the modem configuration file from the main processor or the security processor when present once the encrypted seed is saved as the global variable for the password-of-the-day component. In implementations, the password-of-the-day component is further configured to determine whether a modem configuration file from the modem processor is available on the main processor or the security processor when present, obtain the modem configuration file from the modem processor to the main processor or the security processor when present using an available Application Programming Interface call if the modem configuration file is not on the main processor or the security processor when present, fetch the encrypted seed from the modem configuration file on the main processor or the security processor when present, and store the encrypted seed as a global variable for access by the password-of-the-day component for password-of-the-day generation. In implementations, the modem configuration file is obtained once for a new or updated encrypted seed. In implementations, the password-of-the-day component is further configured to delete the modem configuration file obtained from the main processor or the security processor when present once the encrypted seed is saved as the global variable for the password-of-the-day component. In implementations, the access request is a web based access request. In implementations, the access request is a Hypertext Transfer Protocol Secure (HTTPS) protocol based access request.

Disclosed herein is a method for configuring a main processor or security processor in a customer premises equipment with an encrypted seed for PoTD processing. In implementations, a method for password-of-the-day processing includes receiving, at a main processor via a secondary processor, an access request to an access device from a user device, the main processor and the secondary processor in a multi-processor configuration on the access device, accessing, by a password-of-the-day component on the main processor or an available security processor in response to the access request, an encrypted seed stored on the main processor or the available security processor, respectively, generating, by the password-of-the-day component on the main processor or the available security processor, a password-of-the-day from the encrypted seed, verifying, by the password-of-the-day component on the main processor or the available security processor, a password-of-the-day provided via the user device against the generated password-of-the-day, and transmitting, by the password-of-the-day component on the main processor or the available security processor, a decision to the user device based on verification processing results.

In implementations, a web server configuration file on the main processor is provisioned with the encrypted seed and

15

the accessing further includes fetching the encrypted seed from the web server configuration file. In implementations, the accessing further includes fetching the encrypted seed from a modem configuration file previously obtained from the secondary processor by the main processor or the available security processor, and storing the encrypted seed as a global variable for access by the password-of-the-day component for password-of-the-day generation. In implementations, the accessing further includes getting a modem configuration file from the secondary processor to the main processor or the available security processor using an available Application Programming Interface call, fetching the encrypted seed from the modem configuration file on the main processor or the available security processor, and storing the encrypted seed as a global variable for access by the password-of-the-day component for password-of-the-day generation. In implementations, the accessing further includes starting a timer to limit an access session duration to a configurable period of time when the user gains access to the access device and setting a number of retries the user has to gain access before the user is locked out when the user is denied access to the access device. In implementations, the accessing further includes determining whether a modem configuration file from the secondary processor is available on the main processor or the available security processor, obtaining the modem configuration file from the secondary processor to the main processor or the available security processor using an available Application Programming Interface call if the modem configuration file is not on the main processor or the available security processor, fetching the encrypted seed from the modem configuration file on the main processor or the available security processor, and storing the encrypted seed as a global variable for access by the password-of-the-day component for password-of-the-day generation.

In implementations, an access device includes a primary processor in communication with a secondary processor, and a password-of-the-day component on one of the primary processor or a security processor. In response to an access request received at the access device, the password-of-the-day component is configured to access a locally stored encrypted seed, verify a user provided password-of-the-day with a password-of-the-day generated from the encrypted seed, and send an access response based on results from a verification process to the user device.

In implementations, a web server configuration file on the primary processor is provisioned with the encrypted seed and the password-of-the-day component is configured to fetch the encrypted seed from the web server configuration file. In implementations, the password-of-the-day component is further configured to fetch the encrypted seed from a modem configuration file, the modem configuration file obtained from the secondary processor and stored locally, and store the encrypted seed as a global variable for access by the password-of-the-day component for password-of-the-day generation.

Disclosed herein is a method for configuring a main processor or security processor in a customer premises equipment with an encrypted seed for PoTD processing. A customer premises equipment device includes a modem processor, a main processor in communication with the modem processor; and a password-of-the-day component included on the main processor. The main processor is configured to receive, via the modem processor from a user device, an access request to the customer premises equipment device. In response to the access request, the password-of-the-day component is configured to access an

16

encrypted seed stored on the main processor, generate a password-of-the-day from the encrypted seed, and provide an access response based on comparison of the generated password-of-the-day with a password-of-the-day provided via the user device.

In implementations, a web server configuration file on the main processor is provisioned with the encrypted seed and the password-of-the-day component is configured to fetch the encrypted seed from the web server configuration file. In implementations the password-of-the-day component is further configured to fetch the encrypted seed from a modem configuration file previously obtained from the modem processor by the main processor, and store the encrypted seed as a global variable for access by the password-of-the-day component for password-of-the-day generation. In implementations, the password-of-the-day component is further configured to get a modem configuration file from the modem processor to the main processor using an available Application Programming Interface call when the modem configuration file is unavailable on the main processor, fetch the encrypted seed from the modem configuration file on the main processor, and store the encrypted seed as a global variable for access by the password-of-the-day component for password-of-the-day generation. In implementations, the modem configuration file is obtained once for a new or updated encrypted seed. In implementations, the password-of-the-day component is further configured to delete the modem configuration file from the main processor once the encrypted seed is saved as the global variable for the password-of-the-day component. In implementations, the customer premises equipment device further includes a security processor. The password-of-the-day component is included on the security processor in lieu of on the main processor and the password-of-the-day component on the security processor is configured to access the encrypted seed from the security processor. In implementations, the password-of-the-day component on the security processor is configured to fetch the encrypted seed from a web server configuration file on the main processor. In implementations, the password-of-the-day component on the security processor is configured to fetch the encrypted seed from a modem configuration file previously obtained from the modem processor by the security processor and store the encrypted seed as a global variable for access by the password-of-the-day component for password-of-the-day generation. In implementations, the password-of-the-day component on the security processor is further configured to get a modem configuration file from the modem processor to the security processor using an available Application Programming Interface call when the modem configuration file is not present on the security processor, fetch the encrypted seed from the modem configuration file on the security processor, and store the encrypted seed as a global variable for access by the password-of-the-day component for password-of-the-day generation. In implementations, the access request is a web based access request. In implementations, the access request is a Hypertext Transfer Protocol Secure (HTTPS) protocol based access request.

In implementations, a method for password-of-the-day processing includes receiving, at a main processor via a secondary processor, an access request to an access device from a user device, the main processor and the secondary processor in a multi-processor configuration on the access device, accessing, by a password-of-the-day component on the main processor in response to the access request, an encrypted seed stored on the main processor, generating, by the password-of-the-day component on the main processor,

a password-of-the-day from the encrypted seed, verifying, by the password-of-the-day component on the main processor, a password-of-the-day provided via the user device against the generated password-of-the-day, and transmitting, by the password-of-the-day component on the main processor, a decision to the user device based on verification processing results.

In implementations, a web server configuration file on the main processor is provisioned with the encrypted seed and the accessing further includes fetching the encrypted seed from the web server configuration file. In implementations, the accessing further includes fetching the encrypted seed from a modem configuration file previously obtained from the secondary processor by the main processor, and storing the encrypted seed as a global variable for access by the password-of-the-day component for password-of-the-day generation. In implementations, the accessing further includes getting a modem configuration file from the secondary processor to the main processor using an available Application Programming Interface call when the modem configuration file is unavailable on the main processor, fetching the encrypted seed from the modem configuration file on the main processor, and storing the encrypted seed as a global variable for access by the password-of-the-day component for password-of-the-day generation. In implementations, the accessing further includes starting a timer to limit an access session duration to a configurable period of time when the user device gains access to the access device and setting a number of retries the user device has to gain access before the user device is locked out when the user device is denied access to the access device. In implementations, the encrypted seed is stored on a security processor, wherein the password-of-the-day component is on the security processor, and wherein the accessing accesses the encrypted seed stored on the security processor by the password-of-the-day component on the security processor. In implementations, the accessing further includes fetching the encrypted seed from a web server configuration file on the main processor, wherein the web server configuration file is provisioned with the encrypted seed. In implementations, the accessing further includes fetching the encrypted seed from a modem configuration file previously obtained from the secondary processor by the security processor; and storing the encrypted seed as a global variable for access by the password-of-the-day component for password-of-the-day generation.

In implementations, an access device includes a primary processor in communication with a secondary processor; and a password-of-the-day component on the primary processor, wherein, in response to an access request received at the access device. The password-of-the-day component is configured to access a locally stored encrypted seed, verify a user provided password-of-the-day with a password-of-the-day generated from the encrypted seed, and send an access response based on results from a verification process to a user device.

In implementations, a web server configuration file on the primary processor is provisioned with the encrypted seed and the password-of-the-day component is configured to fetch the encrypted seed from the web server configuration file. In implementations, the password-of-the-day component is further configured to fetch the encrypted seed from a modem configuration file, the modem configuration file obtained from the secondary processor and stored locally, and store the encrypted seed as a global variable for access by the password-of-the-day component for password-of-the-day generation. In implementations, the access device fur-

ther including a security processor, where the password-of-the-day component is included on the security processor in lieu of on the primary processor.

Although some implementations herein refer to methods, it will be appreciated by one skilled in the art that they may also be embodied as a system or computer program product. Accordingly, aspects of the present disclosure may take the form of an entirely hardware implementation, an entirely software implementation (including firmware, resident software, micro-code, etc.) or an implementation combining software and hardware aspects that may all generally be referred to herein as a "processor," "device," or "system." Furthermore, aspects of the present disclosure may take the form of a computer program product embodied in one or more computer readable mediums having the computer readable program code embodied thereon. For example, the computer readable mediums can be non-transitory. Any combination of one or more computer readable mediums may be utilized. The computer readable medium may be a computer readable signal medium or a computer readable storage medium. A computer readable storage medium may be, for example, but not limited to, an electronic, magnetic, optical, electromagnetic, infrared, or semiconductor system, apparatus, or device, or any suitable combination of the foregoing. More specific examples (a non-exhaustive list) of the computer-readable storage medium include the following: an electrical connection having one or more wires, a portable computer diskette, a hard disk, a random access memory (RAM), a read-only memory (ROM), an erasable programmable read-only memory (EPROM or Flash memory), an optical fiber, a portable compact disc read-only memory (CD-ROM), an optical storage device, a magnetic storage device, or any suitable combination of the foregoing. In the context of this document, a computer-readable storage medium may be any tangible medium that can contain, or store a program for use by or in connection with an instruction execution system, apparatus, or device.

A computer readable signal medium may include a propagated data signal with computer readable program code embodied therein, for example, in baseband or as part of a carrier wave. Such a propagated signal may take any of a variety of forms, including, but not limited to, electromagnetic, optical, or any suitable combination thereof. A computer readable signal medium may be any computer readable medium that is not a computer readable storage medium and that can communicate, propagate, or transport a program for use by or in connection with an instruction execution system, apparatus, or device.

Program code embodied on a computer readable medium may be transmitted using any appropriate medium, including but not limited to CDs, DVDs, wireless, wireline, optical fiber cable, RF, etc., or any suitable combination of the foregoing.

Computer program code for carrying out operations for aspects of the present disclosure may be written in any combination of one or more programming languages, including an object-oriented programming language such as Java, Smalltalk, C++ or the like and conventional procedural programming languages, such as the "C" programming language or similar programming languages. The program code may execute entirely on the user's computer, partly on the user's computer, as a stand-alone software package, partly on the user's computer and partly on a remote computer or entirely on the remote computer or server. In the latter scenario, the remote computer may be connected to the user's computer through any type of network, including a local area network (LAN) or a wide area network (WAN), or

the connection may be made to an external computer (for example, through the Internet using an Internet Service Provider).

Aspects of the present disclosure are described herein with reference to flowchart illustrations and/or block diagrams of methods, apparatus (systems) and computer program products according to implementations of the disclosure. It will be understood that each block of the flowchart illustrations and/or block diagrams, and combinations of blocks in the flowchart illustrations and/or block diagrams, can be implemented by computer program instructions.

These computer program instructions may be provided to a processor of a general purpose computer, special purpose computer, or other programmable data processing apparatus to produce a machine, such that the instructions, which execute via the processor of the computer or other programmable data processing apparatus, create means for implementing the functions/acts specified in the flowchart and/or block diagram block or blocks. These computer program instructions may also be stored in a computer readable medium that can direct a computer, other programmable data processing apparatus, or other devices to function in a particular manner, such that the instructions stored in the computer readable medium produce an article of manufacture including instructions which implement the function/act specified in the flowchart and/or block diagram block or blocks.

The computer program instructions may also be loaded onto a computer, other programmable data processing apparatus, or other devices to cause a series of operational steps to be performed on the computer, other programmable apparatus or other devices to produce a computer implemented process such that the instructions which execute on the computer or other programmable apparatus provide processes for implementing the functions/acts specified in the flowchart and/or block diagram block or blocks.

The flowcharts and block diagrams in the figures illustrate the architecture, functionality, and operation of possible implementations of systems, methods, and computer program products according to various implementations of the present disclosure. In this regard, each block in the flowchart or block diagrams may represent a module, segment, or portion of code, which comprises one or more executable instructions for implementing the specified logical function(s). It should also be noted that, in some alternative implementations, the functions noted in the block may occur out of the order noted in the figures.

While the disclosure has been described in connection with certain implementations, it is to be understood that the disclosure is not to be limited to the disclosed implementations but, on the contrary, is intended to cover various modifications, combinations, and equivalent arrangements included within the scope of the appended claims, which scope is to be accorded the broadest interpretation so as to encompass all such modifications and equivalent structures as is permitted under the law.

What is claimed is:

1. A customer premises equipment device, comprising:
a modem processor;

a main processor in communication with the modem processor; and

a password-of-the-day component included on the main processor,

wherein the main processor is configured to receive, via the modem processor from a user device, an access request to the customer premises equipment device, and

wherein, in response to the access request, the password-of-the-day component is configured to:

access an encrypted seed stored on the main processor;
generate a password-of-the-day from the encrypted seed; and

provide an access response based on comparison of the generated password-of-the-day with a password-of-the-day provided via the user device.

2. The customer premises equipment device of claim 1, wherein a web server configuration file on the main processor is provisioned with the encrypted seed and the password-of-the-day component is configured to fetch the encrypted seed from the web server configuration file.

3. The customer premises equipment device of claim 1, wherein the password-of-the-day component is further configured to:

fetch the encrypted seed from a modem configuration file previously obtained from the modem processor by the main processor; and

store the encrypted seed as a global variable for access by the password-of-the-day component for password-of-the-day generation.

4. The customer premises equipment device of claim 1, wherein the password-of-the-day component is further configured to:

get a modem configuration file from the modem processor to the main processor using an available Application Programming Interface call when the modem configuration file is unavailable on the main processor;

fetch the encrypted seed from the modem configuration file on the main processor; and

store the encrypted seed as a global variable for access by the password-of-the-day component for password-of-the-day generation.

5. The customer premises equipment device of claim 4, wherein the modem configuration file is obtained once for a new or updated encrypted seed.

6. The customer premises equipment device of claim 5, wherein the password-of-the-day component is further configured to:

delete the modem configuration file from the main processor once the encrypted seed is saved as the global variable for the password-of-the-day component.

7. The customer premises equipment device of claim 1, further comprising:

a security processor,

wherein the password-of-the-day component is included on the security processor in lieu of on the main processor; and

wherein the password-of-the-day component on the security processor is configured to access the encrypted seed from the security processor.

8. The customer premises equipment device of claim 7, wherein the password-of-the-day component on the security processor is configured to fetch the encrypted seed from a web server configuration file on the main processor.

9. The customer premises equipment device of claim 7, wherein the password-of-the-day component on the security processor is configured to:

fetch the encrypted seed from a modem configuration file previously obtained from the modem processor by the security processor; and

store the encrypted seed as a global variable for access by the password-of-the-day component for password-of-the-day generation.

21

10. The customer premises equipment device of claim 7, wherein the password-of-the-day component on the security processor is further configured to:

- get a modem configuration file from the modem processor to the security processor using an available Application Programming Interface call when the modem configuration file is not present on the security processor;
- fetch the encrypted seed from the modem configuration file on the security processor; and
- store the encrypted seed as a global variable for access by the password-of-the-day component for password-of-the-day generation.

11. The customer premises equipment device of claim 1, wherein the access request is a web based access request.

12. The customer premises equipment device of claim 1, wherein the access request is a Hypertext Transfer Protocol Secure (HTTPS) protocol based access request.

13. A method for password-of-the-day processing, the method comprising:

- receiving, at a main processor via a secondary processor, an access request to an access device from a user device, the main processor and the secondary processor in a multi-processor configuration on the access device;
- accessing, by a password-of-the-day component on the main processor in response to the access request, an encrypted seed stored on the main processor;
- generating, by the password-of-the-day component on the main processor, a password-of-the-day from the encrypted seed;
- verifying, by the password-of-the-day component on the main processor, a password-of-the-day provided via the user device against the generated password-of-the-day; and
- transmitting, by the password-of-the-day component on the main processor, a decision to the user device based on verification processing results.

14. The method of claim 13, wherein a web server configuration file on the main processor is provisioned with the encrypted seed and the accessing further comprises:

- fetching the encrypted seed from the web server configuration file.

15. The method of claim 13, wherein the accessing further comprises:

- fetching the encrypted seed from a modem configuration file previously obtained from the secondary processor by the main processor; and
- storing the encrypted seed as a global variable for access by the password-of-the-day component for password-of-the-day generation.

16. The method of claim 13, wherein the accessing further comprises:

- getting a modem configuration file from the secondary processor to the main processor using an available Application Programming Interface call when the modem configuration file is unavailable on the main processor;
- fetching the encrypted seed from the modem configuration file on the main processor; and
- storing the encrypted seed as a global variable for access by the password-of-the-day component for password-of-the-day generation.

17. The method of claim 13, wherein the accessing further comprises:

- starting a timer to limit an access session duration to a configurable period of time when the user device gains access to the access device; and

22

setting a number of retries the user device has to gain access before the user device is locked out when the user device is denied access to the access device.

18. The method of claim 13, wherein the encrypted seed is stored on a security processor, wherein the password-of-the-day component is on the security processor, and wherein the accessing accesses the encrypted seed stored on the security processor by the password-of-the-day component on the security processor.

19. The method of claim 18, wherein the accessing further comprises:

- fetching the encrypted seed from a web server configuration file on the main processor, wherein the web server configuration file is provisioned with the encrypted seed.

20. The method of claim 18, wherein the accessing further comprises:

- fetching the encrypted seed from a modem configuration file previously obtained from the secondary processor by the security processor; and
- storing the encrypted seed as a global variable for access by the password-of-the-day component for password-of-the-day generation.

21. An access device, comprising:

- a primary processor in communication with a secondary processor, wherein a web server configuration file on the primary processor is provisioned with an encrypted seed; and

- a password-of-the-day component on the primary processor, wherein, in response to an access request received at the access device, the password-of-the-day component is configured to:

- fetch the encrypted seed from the web server configuration file;
- verify a user provided password-of-the-day with a password-of-the-day generated from the encrypted seed; and
- send an access response based on results from a verification process to a user device.

22. The access device of claim 21, further comprising:

- a security processor, wherein the password-of-the-day component is included on the security processor in lieu of on the primary processor.

23. An access device, comprising:

- a primary processor in communication with a secondary processor; and

- a password-of-the-day component on the primary processor, wherein, in response to an access request received at the access device, the password-of-the-day component is configured to:

- fetch an encrypted seed from a modem configuration file, the modem configuration file obtained from the secondary processor and stored locally;
- verify a user provided password-of-the-day with a password-of-the-day generated from the encrypted seed; and
- send an access response based on results from a verification process to a user device.

24. The access device of claim 23, wherein the password-of-the-day component is further configured to:

- store the encrypted seed as a global variable for access by the password-of-the-day component for password-of-the-day generation.